INCH-POUND
MIL-PRF-22885/53C
19 February 1993
SUPERSEDING
MIL-S-22885/53B
22 November 1982

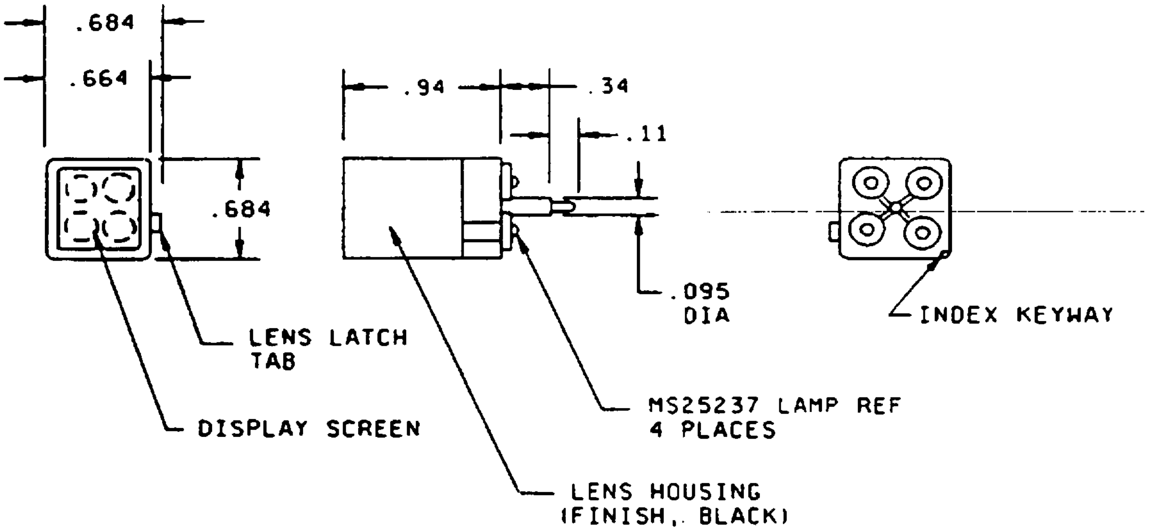
NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

PERFORMANCE SPECIFICATION SHEET

Ⓒ SWITCH, PUSH BUTTON, ILLUMINATED, PUSHBUTTON FOR (WITH LAMP-CAP COLOR FILTERS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the switches described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-22885.



Inches	mm
.095	2.41
.11	2.79
.34	8.64
.664	16.87
.684	17.37
.94	23.88

NOTES:

1. Dimensions are in inches. Metric equivalents are given for general information only.
2. Unless otherwise specified, tolerances are ±.005 (.13 mm) for three place decimals and ±.020 (.51 mm) for two place decimals.
3. Dashed circles represent color filters.

FIGURE 1. Design and construction.

Ⓒ denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

REQUIREMENTS

Ⓒ Design and construction See figure 1 Pushbutton shall accommodate four midget flange base T-1 ¾ incandescent lamps and shall be capable of mating with switch housings in accordance with MIL-S-22885/56 Pushbutton shall accommodate color filters in accordance with MIL-S-22885/55 Unless otherwise specified in contract or purchase order, lamps are not supplied

Enclosure design 1 (Unsealed)

Temperature characteristic 1 (-55°C to +85°C)

Vibration grade 2 (10 - 500 Hz)

Display type W (White until illuminated, then background appears in color)

Material

Display screen Shall be of clear transparent material

Lamp module Lamp insulator block may be fabricated from thermoplastic material

Ⓒ Color and luminance See table I

Part or Identifying Number (PIN) M22885/53-(dash number from table II)

Ⓒ Qualification inspection As a part of illuminated push button switch assembly, MIL-S-22885/56

TABLE I Chromaticity limits and photometric brightness

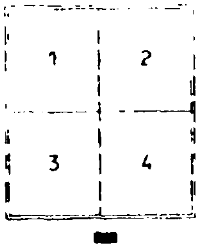
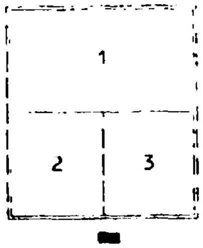
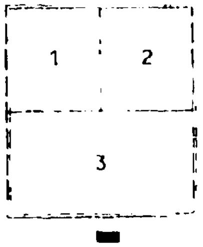
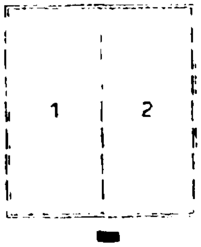
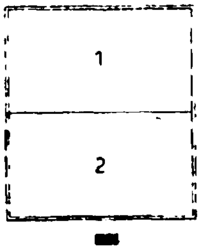
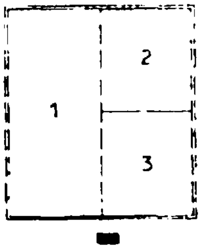
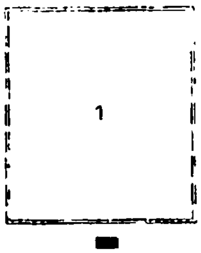
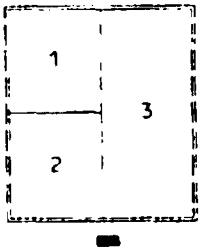
Color	Color code	Chromaticity limits 1/ 2360° Kelvin		Switch photometric brightness (foot-lamberts) 2/ minimum
		x	y	
Ⓒ Blue	B	210	280	25
		330	280	
		330	420	
		210	420	
Green	G	260	570	50
		300	630	
		160	660	
		200	720	
Red	R	695	285	80
		703	SL 3/	
		655	325	
		660	SL 3/	
White	W	380	350	150
		460	.385	
		460	430	
		380	410	
Yellow	Y	562	415	150
		570	SL 3/	
		596	382	
		605	SL 3/	

1/ The chromaticity of the illuminated display, expressed as x and y coordinates on the CIE chromaticity diagram, shall lie within areas bounded by the coordinates listed for each color, using a source illuminate of 2360°K

2/ Switch photometric brightness, when viewed from the front of the applicable pushbutton display and using midget flange, T-1 ¾ incandescent lamps operated at a 34 ± 02 MSCP

3/ SL = Spectrum Locus (where intersected by other coordinate pair)

Table II. [illegible]

Dash number 1/ 2/	Display 3/	Dash number 1/ 2/	Display 3/
-1XXXX		-5XXX	
-2XXX		-6XX	
-3XX		-7XXX	
-4X		-8XXX	

1/ Dash numbers are shown for lamp capsules without color filters. Replace X's with the color codes of table I in the sequence shown.

(C) 2/ Dash numbers as shown replace and are interchangeable with previous dash numbers which included additional zeros (Example -001XXXX).

(C) 3/ Pushbutton assembly removal tab is shown for orientation.

CONCLUDING MATERIAL

Custodians:
  Army - ER
  Navy - EC
  Air Force - 85

Review activities:
  Army - MI
  Navy - SH
  Air Force - 99
  DLA - ES

User activities:
  Army - AP, ME
  Navy - AS, MC, OS
  Air Force - 19

Preparing activity:
  Navy - EC

Agent:
  DLA - ES

(Project 5930-1502-01)